

RHCE

CAPSTONE PROJECT

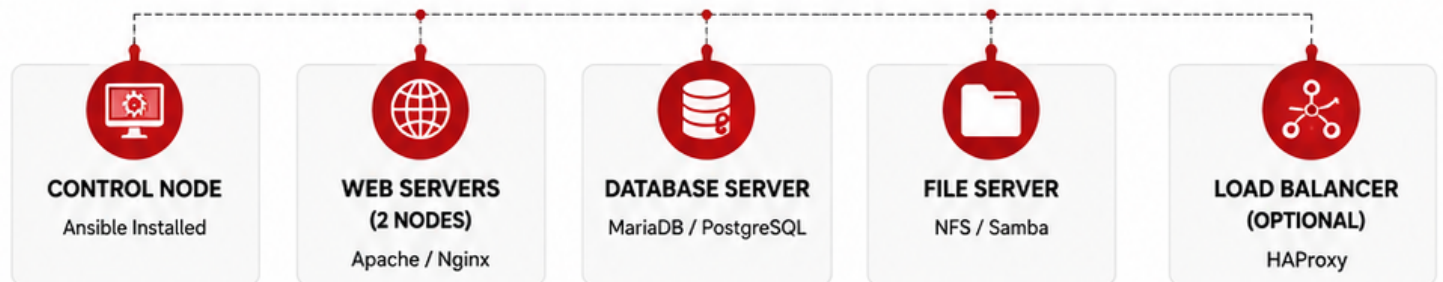
AUTOMATED ENTERPRISE LINUX INFRASTRUCTURE DEPLOYMENT

Design, build, and automate a secure and scalable enterprise infrastructure using Red Hat Enterprise Linux and Ansible.



PROJECT OVERVIEW

You are a system engineer tasked with setting up a complete infrastructure for a web application. Automate everything from server provisioning to security hardening and service deployment.



KEY TASKS



 DELIVERABLES	 SKILLS COVERED	 EVALUATION CRITERIA	 BONUS CHALLENGES										
✓ Ansible Project Directory (inventory, roles, playbooks) ✓ Documentation (Architecture & Setup) ✓ Demo: Full Automated Deployment	• Ansible Automation • RHEL System Administration • Networking & Services • Security Best Practices • Troubleshooting	<table><tr><td>Automation Completeness</td><td>40%</td></tr><tr><td>Correct Configuration</td><td>25%</td></tr><tr><td>Security Implementation</td><td>15%</td></tr><tr><td>Documentation</td><td>10%</td></tr><tr><td>Bonus Features</td><td>10%</td></tr></table>	Automation Completeness	40%	Correct Configuration	25%	Security Implementation	15%	Documentation	10%	Bonus Features	10%	• Ansible Vault for Secrets • CI/CD Pipeline • Containerize with Podman • Auto-Scaling • Centralized Logging (ELK Stack)
Automation Completeness	40%												
Correct Configuration	25%												
Security Implementation	15%												
Documentation	10%												
Bonus Features	10%												

AUTOMATE. SECURE. DEPLOY. SUCCEED.

Build like a **Pro**. Automate like an **Engineer**. Think like a **DevOps**.

RHCE Capstone Project

“Automated Enterprise Linux Infrastructure Deployment”

Objective

Design and deploy a fully automated, secure, and scalable Linux environment using Red Hat Enterprise Linux and Ansible.

Scenario

You are a system engineer in a company setting up infrastructure for a web application. You must automate everything—from server provisioning to security hardening and service deployment.

Infrastructure Setup

Minimum 4–6 virtual machines:

- Control Node
 - Ansible installed
- Web Servers (2 nodes)
 - Apache/Nginx
- Database Server
 - MariaDB/PostgreSQL
- File Server
 - NFS or Samba
- *(Optional)* Load Balancer Node

Core Tasks

1. Ansible Automation

- Create inventory files (static + dynamic)
- Write playbooks for:
 - Package installation

- Service configuration
 - User management
- Use:
 - Roles
 - Handlers
 - Variables
 - Templates (Jinja2)

2. Web Server Setup

- Install and configure Apache HTTP Server
- Deploy a sample website
- Configure virtual hosts
- Enable HTTPS (self-signed or Let's Encrypt)

3. Database Server

- Install MariaDB
- Create:
 - Database
 - Users with privileges
- Secure DB (remove test DB, anonymous users)

4. Network File Sharing

- Configure NFS or Samba
- Mount shares on web servers
- Automate mounts via Ansible

5. Security Hardening

- Configure SELinux
- Setup firewall using firewalld
- Implement SSH key-based authentication
- Disable root login

6. Load Balancing (Optional but valuable)

- Configure HAProxy
- Distribute traffic across web servers

7. Monitoring & Logging

- Configure basic logging (rsyslog)
- Optional:
 - Setup Nagios or Grafana/Prometheus

8. Cron Jobs & Maintenance

- Automate backups:
 - Website files
 - Database dumps
- Schedule with cron

Deliverables

Students should submit:

- ✓ Ansible project directory:
 - inventory/
 - roles/
 - playbooks/
- ✓ Documentation:
 - Architecture diagram
 - Setup instructions
- ✓ Demo:
 - Run playbook → full system deploys automatically

Evaluation Criteria

- Automation completeness (40%)
- Correct configuration (25%)
- Security implementation (15%)
- Documentation (10%)
- Bonus features (10%)

SUBMISSION DETAILS

 Submit to: shyam.s@ipsrsolutions.com

 Last Date: 24 May 2026

 Time: 6:00 PM IST

 Late submissions will not be accepted.